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Abstract –

The rapid growth of EV adoption has put real strain on charging infrastructure. Grid-dependent stations drive load instability, raise operational costs, and struggle to absorb renewable energy — largely because most deployed systems lack coordinated real-time monitoring and energy management. Proposed smart-charging solutions have generally stayed in simulation.

This paper presents a hardware-validated IoT architecture for hybrid energy management at EV charging stations. At its centre is a Point of Common Coupling (PCC) that ties together solar PV generation, battery storage, and the utility grid, controlled by a rule-based algorithm that prioritizes renewables. A five-layer framework — spanning sensor networks, edge processing, cloud analytics, and user dashboards — supports real-time monitoring, predictive control, and remote station management. The system was validated through MATLAB/Simulink simulation and a working prototype built on microcontrollers, power converters, and a full sensor suite, with data piped through Node-RED dashboard. Experimental results show measurable improvements in energy utilization, operating cost, and grid stability. The prototype demonstrates deployability without utility-side infrastructure changes.

Key terms — IoT Architecture, EV Charging Infrastructure, Point of Common Coupling (PCC), Solar MPPT, ESS Management, Reverse Grid Integration, MATLAB Simulation, Node-RED Dashboard, Hardware Prototype, Multi-Mode Automation.

I. INTRODUCTION

The transportation sector is electrifying faster than the charging infrastructure can keep up. Global EV sales crossed 10 million units in 2022 and are projected to represent over 60% of new vehicle sales by 2030 [REF]. The bottleneck is no longer the vehicle — it is the station that powers it. Most deployed charging infrastructure today draws entirely from the utility grid, which creates three compounding problems - it adds uncontrolled peak demand to already-stressed distribution networks, exposes operators to volatile grid tariffs, and squanders co-located solar generation that could otherwise reduce both cost and carbon footprint.

The solution direction is not disputed. Pairing solar PV and battery storage with EV charging stations reduces grid draw, stabilizes operating costs, and makes the environmental case for EVs substantially stronger. There is largely unproven is whether such systems can be built, system integrated, and operated reliably outside a simulation environment. A survey of recent literature reveals a consistent pattern, architectures are proposed, energy flows are modelled in MATLAB, efficiency figures are reported, and the hardware is never built. The gap between a promising simulation and a deployable station has remained largely unclosed.

The energy management logic at its core is a deterministic five-state PCC automation algorithm that responds to real-time solar availability and ESS state-of-charge thresholds, governing dispatch across three sources — solar PV, ESS, and the utility grid. When ESS charge exceeds defined upper thresholds during active solar generation, surplus energy is returned to the grid, converting the station from a net consumer to a prosumer. These dispatch behaviours are not simulation targets — they are measured outcomes from a working hardware prototype.

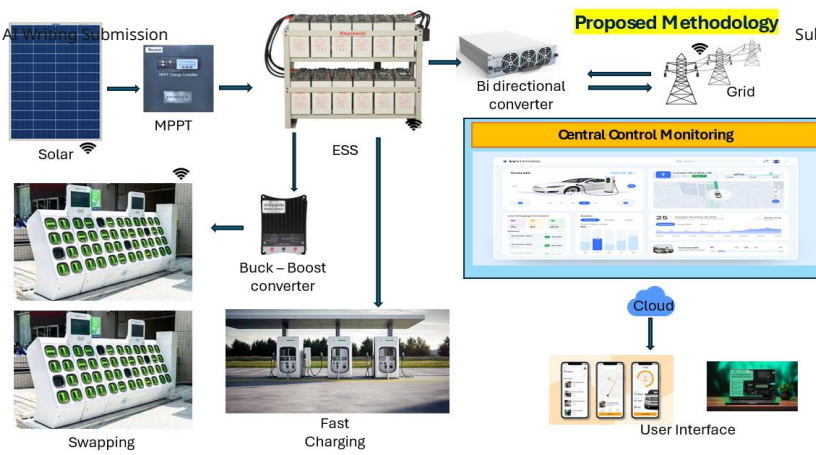


Fig: 1. High-level system overview of the IoT-enabled hybrid EV charging architecture.

The system simultaneously manages two operationally distinct load types - a Battery Swapping Station (BSS) and DC Charging ports. As per-vehicle charging parameters — current setpoints, charging duration, and energy delivered — are computed dynamically from each vehicle's battery capacity, present SOC, and rated voltage. This dual-load architecture has not been previously validated in hardware under a unified IoT management framework, and it represents a meaningful step beyond the single-load topologies that dominate the literature. The supervisory architecture is a complete five-layer IoT stack. The Device Layer handles physical sensing and actuation via ESP32 microcontrollers. The Network Layer uses MQTT for low-latency telemetry. The Edge Computing Layer runs MPPT and charge-cycle algorithms locally to eliminate cloud-round-trip latency during source transitions. The Application Layer delivers a live Node-RED dashboard over HTTPS. All five layers are operational in the prototype, not just in the architecture diagram. The validation follows a two-stage methodology. MATLAB/Simulink simulations first established energy flow behaviour and confirmed control logic across varied irradiance and load profiles. A bench-level hardware prototype then replicated those conditions with production-grade power converters, a full sensor suite covering voltage, current, and temperature, and lead-acid batteries representing the ESS. The close agreement between simulated energy flows and measured prototype data confirms both the fidelity of the model and the practical realizability of the system. The architecture requires no utility-side infrastructure changes and is modular enough to extend to commercial buildings and industrial facilities sharing the same solar-ESS-grid source configuration.

II. LITERATURE SURVEY

The growing deployment of EVs has generated substantial research into smart charging infrastructure, with foundational studies [1]–[3] establishing the grid-side consequences of uncoordinated charging—demand instability, voltage deviation, and the carbon overhead of fossil-fuel-dependent supply. IoT-based monitoring and predictive scheduling frameworks [9], [10] demonstrated the value of data-driven control, yet these architectures remained narrowly scoped, addressing residential interfaces [17] or isolated battery monitoring [10] without a unified framework spanning generation, storage, and load. Parallel work on PV-predictive ESS management [14] and microgrid optimization [15] showed meaningful reductions in grid dependency but operated entirely within simulation environments, leaving unresolved the practical constraints of sensor interoperability, communication protocol selection, and edge-to-cloud pipeline reliability. Grid-support studies on active power filters [18] and fault detection [19] addressed legitimate power quality concerns but were never integrated into a dynamic source-switching strategy at the Point of Common Coupling. Battery swapping research [22]–[24] made a strong operational case for reduced vehicle downtime and smoother grid demand, yet consistently treated swapping stations as standalone systems, separate from fast-charging infrastructure and the renewable assets that could serve both.

The Hybrid renewable energy management, PCC-based automation, dual-load control across battery swapping and DC fast charging stations, and a complete cloud-edge IoT stack have each been studied in isolation—none have been combined, and none have been validated in hardware. The present work addresses this directly by delivering a prototype validated five-layer IoT architecture that physically demonstrates coordinated dispatch across solar PV, ESS, and grid sources through an automated PCC, while managing both load types under live cloud supervision.

III. METHODOLOGY

This section describes the design, simulation, and hardware implementation of the proposed IoT-enabled hybrid energy management system. The methodology is organized around four interrelated pillars: the five-layer IoT architecture governing system coordination and data flow; the power flow management and operational logic at the Point of Common Coupling; the core automation and control algorithms; and the hardware prototyping and validation process. Together, these pillars trace a deliberate path from conceptual design through MATLAB/Simulink simulation to a fully operational bench prototype.

A. Proposed System Design and Architecture

The system is built around a hierarchical five-layer IoT architecture that handles data acquisition, local processing, cloud communication, and user supervision as distinct but tightly coupled functions. Each layer has a defined responsibility and a defined interface with the layers above and below it, which is what separates this architecture from the loosely coupled monitoring systems common in prior work.

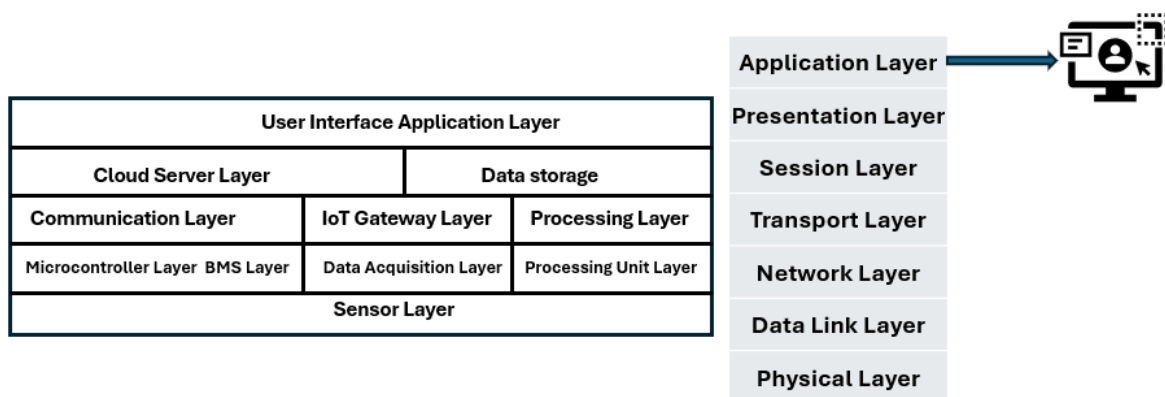


Fig: 2. Proposed 5 layered IoT Architecture.

The lowest layer is the Sensor Layer, which forms the physical interface between the IoT stack and the electrical infrastructure of the charging station. Voltage and current measurements are taken at three points — the solar PV output terminals, the ESS battery bank, and the grid connection — using ACS712 Hall-effect modules selected for their linearity and galvanic isolation. Temperature and humidity at the ESS enclosure and charging units are monitored via DHT11 sensors, and insulation resistance detectors provide a continuous ground-fault indicator. Every control and monitoring decision in the layers above depends on the accuracy and update rate of this layer, so sensor placement and calibration were treated as primary design decisions rather than afterthoughts.

The second layer is the Microcontroller and Data Acquisition Layer, which provides local edge intelligence. Two ESP32 development boards divide the edge workload. The first handles data aggregation — collecting raw analog and digital streams from all sensors, applying signal conditioning including filtering and scaling, and packaging telemetry for upstream transmission. The second runs time-sensitive control algorithms locally: SOC estimation from terminal voltage and current integration, MPPT execution for the solar converter, and real-time threshold comparison for PCC state transitions. Splitting acquisition and control across two devices prevents processing contention during high-frequency switching events and improves determinism in the control loop — a design choice validated during prototype testing.

The Communication Layer sits above the edge devices and is responsible for reliable data transport to the cloud backend. A dual-protocol strategy was adopted based on the nature of each data flow. MQTT handles continuous telemetry streaming from both ESP32 units to the AWS IoT Core broker, chosen for its low overhead and publish-subscribe model that decouples sensor update rates from cloud ingestion rates. HTTP/HTTPS handles RESTful API calls for dashboard configuration updates and command delivery from the cloud to the edge, where the request-response model is more appropriate than a persistent subscription.

The energy management strategy governs power exchange between three sources — solar PV, the Energy Storage System (ESS), and the utility grid — and two load types — Battery Swapping Stations (BSS) and DC Fast Chargers (DCFC) — at the Point of Common Coupling (PCC). Rather than treating the PCC as a passive junction, the proposed architecture positions it as an active decision point. A state-machine-based control algorithm running on the cloud supervisory layer continuously evaluates three real-time inputs: solar generation availability, ESS state of charge, and active load demand. Based on these inputs, the system transitions between five discrete operating modes, each representing a distinct power flow configuration. The mode definitions and their transition conditions are described below and illustrated in Fig. 3.

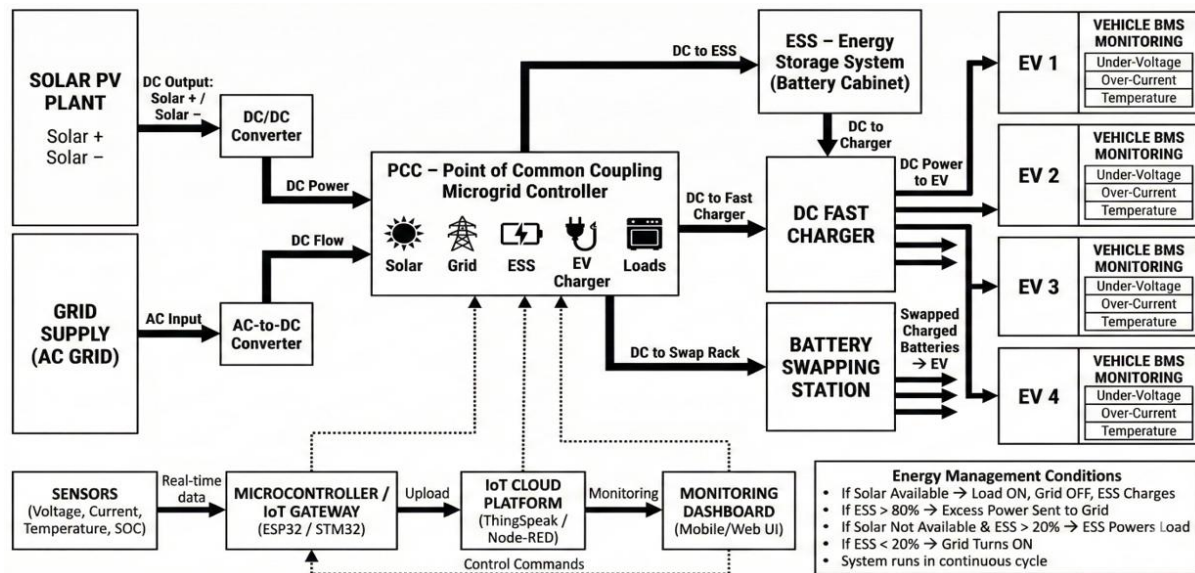


Fig. 3. Block diagram of the IoT-enabled hybrid EV charging system

In Mode 1 (Solar Priority), solar generation is active and ESS charge is below the upper threshold. Available PV power simultaneously serves the connected loads and charges the ESS. Grid connection is held open but draws zero current. This is the preferred steady-state operating condition — the system remains in this mode as long as irradiance is sufficient to cover load demand, and grid exposure is effectively zero.

When solar generation drops below the minimum generation threshold, the system transitions to Mode 2 (ESS Support). The ESS discharges to meet load demand and the grid remains disconnected. This mode preserves grid independence during cloud cover or transient shading events, drawing down stored energy before resorting to utility power. The ESS lower threshold — set at 20% SOC — defines the floor below which this mode can no longer sustain the load.

Mode 3 (Grid Backup) activates when both solar generation is unavailable and ESS SOC falls below 20%. The grid connects and supplies the full load. This mode is designed as a fallback condition rather than a normal operating state — its activation frequency in measured data serves as a practical indicator of system sizing adequacy, and minimizing its duration is a direct design objective.

Mode 4 (Reverse Grid Operation) handles the surplus generation condition. When solar is active and ESS SOC exceeds 80%, continued charging would waste available generation. Instead, a bidirectional converter reverses power flow direction and exports surplus energy to the grid. This transitions the station from a net consumer to a grid prosumer — a capability that is architecturally supported by the PCC design and experimentally demonstrated in the prototype, distinguishing this work from systems that treat reverse power flow as a theoretical option rather than an implemented function.

Mode 5 (ESS Grid Charging) addresses the overnight or low-demand recovery scenario. When solar is unavailable and the ESS is depleted, the grid charges the ESS with load disconnected. This mode is intentionally restricted to off-peak electricity windows to minimize tariff exposure, and the charge cycle is terminated automatically when ESS

C. PCC Integration and Automation

The Point of Common Coupling is the physical and logical convergence point where solar PV output, ESS terminals, and the utility grid connection meet before power is dispatched to the BSS and DCFC loads. Its design in this system goes beyond passive bus integration — the PCC is an actively controlled switching fabric whose state at any moment determines which sources are connected, which are isolated, and in which direction power flows.

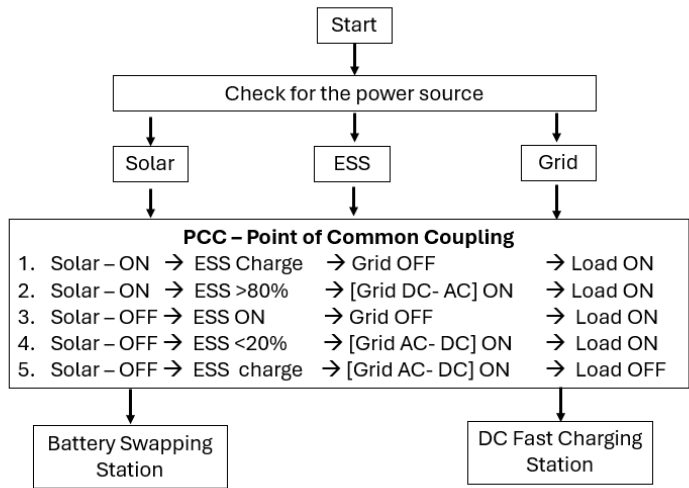


Fig: 4. Flow of 5 methodology in PCC

At the hardware level, source-to-bus connections are managed through a combination of solid-state relays (SSRs) and electromechanical contactors. SSRs handle high-frequency switching paths where contact wear and switching latency would compromise reliability, while contactors manage the heavier current paths where their lower on-state resistance is advantageous. Both are driven directly from the GPIO pins of the ESP32 control board, giving the firmware deterministic control over connection state with no intermediate relay driver abstraction that could introduce timing uncertainty.

The automation logic runs as an embedded C++ state machine on the ESP32 control unit. On each execution cycle, the firmware reads the current sensor outputs — solar generation level, ESS SOC, and load demand — and evaluates them against the threshold conditions that define each operating mode. When a transition condition is met, the firmware issues the corresponding GPIO commands to open or close the appropriate relays, completing the source reconfiguration within one control cycle. This edge-level execution ensures that mode transitions respond to real-time electrical conditions without waiting for a cloud round-trip, which is critical during events such as sudden irradiance drops or ESS threshold crossings where delayed switching could cause momentary supply interruption.

Cloud-level supervisory control operates in parallel as an oversight layer rather than a primary control path. The ESP32 publishes its current mode, sensor readings, and relay states to AWS IoT Core on each telemetry cycle. The cloud layer monitors for anomalous conditions — sustained grid backup activation, abnormal ESS discharge rates, or sensor dropout — and can issue override commands back to the edge when operator intervention is warranted.

This two-tier control structure, where edge firmware handles real-time switching and the cloud handles supervision and exception management, is a deliberate architectural choice that keeps the system responsive under normal conditions while preserving remote override capability. The decision flow and state transition sequence are illustrated in Fig. 5.

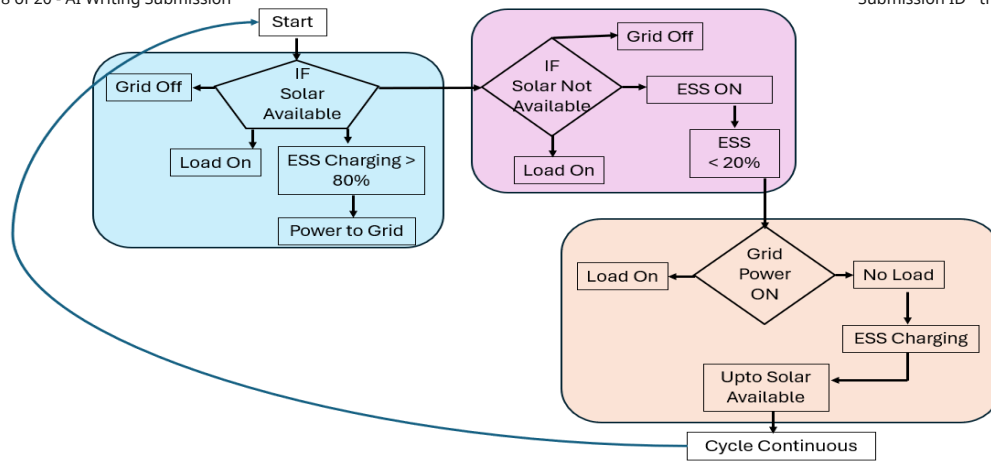


Fig: 5. PCC Automation Logic Flowchart.

D. Core Algorithms

The intelligent operation of the hybrid energy management system depends on four coordinated algorithms distributed across the edge and cloud layers. Each algorithm addresses a distinct aspect of system control — energy state estimation, power electronics regulation, solar harvesting, and supervisory source dispatch — and their outputs feed into one another in a defined execution hierarchy. Together they form the computational backbone that makes autonomous PCC operation possible.

1. The SOC Estimation accurate knowledge of ESS remaining capacity is the prerequisite for every source-switching decision in the system. A Coulomb-counting method is used as the primary estimation technique, integrating measured charge and discharge current over time to track capacity consumed and replenished. Coulomb counting accumulates error over extended operation due to sensor offset and integration drift, so the estimate is periodically corrected against open-circuit voltage measurements taken during rest intervals, using a pre-characterized OCV–SOC lookup table for the lead-acid cells used in the prototype. The corrected SOC percentage is the input that triggers mode transitions at the 20% lower threshold and the 80% upper threshold defined in the PCC state machine.
2. Once the SOC estimate is available, the ESS power electronics are regulated through a two-phase charging profile and a controlled discharge strategy. During charging, the algorithm initiates a constant-current phase when SOC is below the upper threshold, delivering maximum charge current while cell voltage remains within safe limits. As the ESS approaches full capacity, the algorithm transitions to constant-voltage regulation, tapering current to prevent overcharge while completing the charge cycle. During discharge, output current is regulated to maintain a controlled draw rate that avoids deep discharge stress on the cells. Both phases are executed by the ESP32 control board through PWM duty cycle commands to the DC-DC converter, with cycle-by-cycle current feedback closing the regulation loop.
3. Solar PV output varies continuously with irradiance intensity, cell temperature, and partial shading, and a fixed operating point would leave substantial generation unrealized. A Perturb and Observe (P&O) algorithm is implemented on the edge ESP32 to track the maximum power point of the PV array in real time. On each execution step, the algorithm applies a small perturbation to the duty cycle of the DC-DC boost converter and measures the resulting change in output power. If power increases, the perturbation direction is maintained; if power decreases, the direction reverses. This iterative process converges the operating point to the vicinity of the maximum power point under any irradiance condition, with step size tuned to balance tracking speed against steady-state oscillation around the MPP. MPPT execution runs at the edge rather than the cloud to ensure that converter duty cycle adjustments track irradiance changes without communication latency.
4. The supervisory algorithm integrates the outputs of the three subsystems above — current SOC estimate, ESS charge/discharge status, and MPPT-reported solar availability — along with real-time load demand from the sensor layer, to determine which of the five operating modes is currently appropriate. On each evaluation cycle, the algorithm compares the aggregated inputs against the mode transition conditions defined in Section III-B. When a transition is warranted, it generates the corresponding relay switching commands and issues them to the PCC

hardware via the ESP32 GPIO interface. When no transition is warranted, it confirms the current mode and logs the system state to the cloud. This algorithm is the integration point where sensor data, energy state, and control action converge into a single autonomous decision — and it is the mechanism that allows the system to operate through a full day of varying solar conditions and load events without operator input.

IV. Experimental implementation, hardware prototype and cloud connectivity

The control architecture and algorithms described in Section IV were physically realized as a laboratory-scale prototype designed to validate end-to-end system behaviour under real operating conditions. The prototype, shown in Fig. 6, is not a partial bench assembly — it is a complete, integrated implementation of all five IoT layers, from physical sensors through to live cloud dashboards, operating simultaneously on a single hardware platform. Every mode transition described in the state machine was tested on this hardware, and every data point in the results section was measured from it.



Fig: 6 Hardware Development and Prototype.

A. Hardware Prototype Development

The simulation-validated architecture was physically implemented as a laboratory-scale prototype operating at scaled-down power levels, with all functional relationships from the simulation model preserved in hardware. The complete experimental setup is shown in Fig. 6. The Edge intelligence is distributed across two ESP32 DevKit V1 boards with distinct, non-overlapping responsibilities. The primary ESP32 handles all data acquisition — reading ACS712 Hall-effect current sensors at the solar simulator output, ESS terminals, and grid connection point; receiving scaled voltage readings from resistor-divider networks across each source and the DC bus; monitoring ambient temperature at the ESS enclosure via DHT11; and reading load request states from GPIO-connected switches representing the BSS and DCFC ports. All sensor readings undergo filtering and calibration on the primary board before being packetized and forwarded over a serial UART link to the secondary unit.

The secondary ESP32 handles control and communication exclusively. It receives processed data packets from the primary board, executes the SOC estimation algorithm and PCC mode evaluation logic, and issues GPIO switching commands to the solid-state relay modules that physically connect and isolate sources at the PCC. Separating acquisition from control across two processors eliminates the scheduling contention that would arise if sensor polling and relay switching competed for the same execution context — a design choice confirmed during prototype testing to improve transition timing determinism. The secondary board also manages Wi-Fi connectivity and maintains the persistent MQTT link to the cloud backend.

B. Node-RED Connectivity and Dashboard Monitoring

Real-time visualization, remote monitoring, and data logging are handled through a Node-RED dashboard integrated with the cloud backend via MQTT. The control ESP32 publishes structured JSON packets containing all sensor readings and system state variables to a designated MQTT topic. The system was validated first against a local Mosquitto broker, then migrated to AWS IoT Core for scalable cloud deployment. A Node-RED instance subscribes to the topic, parses incoming JSON payloads, applies unit conversions, and routes processed values to the dashboard visualization nodes.

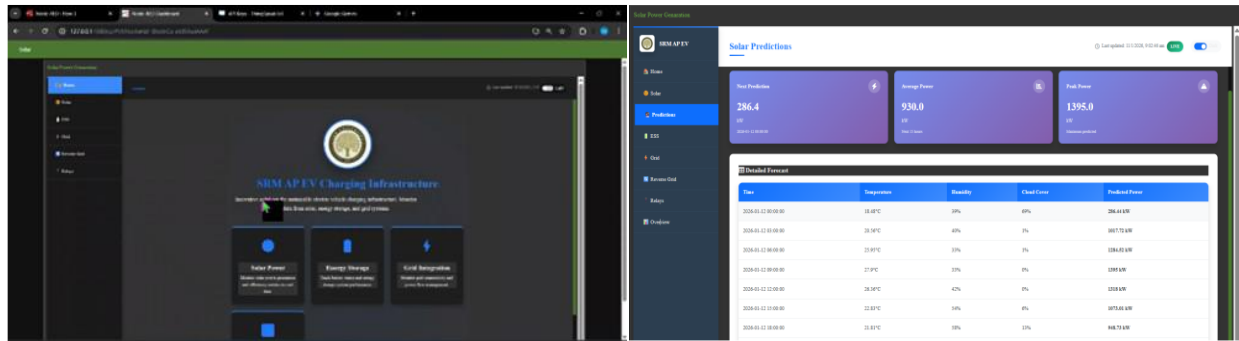


Fig: 7 Node-RED Real-Time Monitoring and Control Dashboard

The dashboard, shown in Fig. 20, presents four synchronized panels covering the complete operational picture of the station. A live power flow diagram updates in real time, displaying active energy paths between sources and loads with numerical voltage, current, and power values at each node. Time-series charts plot solar generation, ESS SOC, grid import and export power, and load demand over operator-selectable windows from minutes to hours. Status indicators show the instantaneous state of each source, the currently active PCC operating mode, and system alarms including over-temperature and critically low SOC conditions. A secure control panel allows authorized operators to issue manual override commands — forcing a specific PCC mode or initiating a controlled ESS charge or discharge test — without requiring direct hardware access.

V. MATLAB/SIMULINK SIMULATION AND RESULTS

The proposed hybrid energy management system was modelled in MATLAB/Simulink to verify control logic, tune algorithm parameters, and establish predicted performance benchmarks against which prototype measurements could later be compared. The simulation environment functioned as a digital twin of the physical system — not as a standalone study, but as the first stage of a two-stage validation methodology whose second stage is the hardware prototype described in Section IV. Results from both stages are compared in Section VI.

A. Overall System Integration and Power Flow Simulation

The top-level Simulink model integrates the three energy sources — a 30 kW solar PV array, a 480V/140A ESS bank, and a 430V grid interface — with the PCC switching logic and downstream loads representing the Battery Swapping Station and DC Fast Charger banks, as illustrated in Fig. 8. Component parameters were taken directly from the hardware prototype specifications so that simulation and physical implementation operate on an identical electrical basis, making the subsequent comparison between predicted and measured results valid rather than approximate.

The PCC automation logic was implemented in Stateflow, with each of the five operating modes from Section III-B represented as a discrete state. State transitions fire on the same threshold conditions used in the embedded firmware — solar availability, ESS SOC relative to the 20% lower and 80% upper bounds, and active load demand — so the simulation and hardware execute the same decision logic from the same inputs. Power electronic switches at the PCC are modelled as ideal switching elements representing the DC-DC boost converters and relay contacts of the physical system, isolating control behaviour from converter parasitic, which are characterized in separate subsystem blocks.

Simulation runs across the full range of operating scenarios — sustained solar generation, cloud-cover transients, ESS depletion, surplus generation with reverse grid export, and overnight ESS recharge — confirmed that mode transitions are smooth and deterministic. No voltage overshoot or current spike was observed at the PCC during any source switching event. Load voltage was maintained continuously through each relay switchover, and the ESS SOC profile followed the expected charge and discharge trajectory under each mode. These results validate the power flow control strategy and confirm that the automation logic performs correctly across the complete operational envelope before any hardware was assembled.

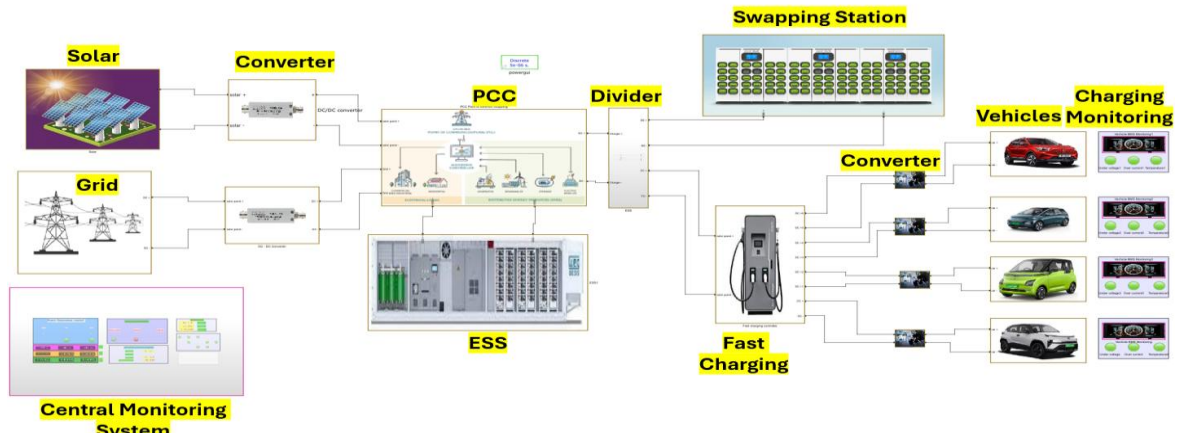


Fig. 8. MATLAB/Simulink Model of the Integrated Hybrid Charging System

B. Solar Generation Unit with MPPT

The solar generation subsystem was modelled using a Simulink PV array block whose output characteristics respond to time-varying irradiance and temperature profiles, replicating the dynamic operating conditions the physical panel would encounter across a typical generation day. A DC-DC boost converter interfaces the PV array output to the 430V DC bus at the PCC, with its duty cycle governed in real time by the Perturb and Observe MPPT algorithm described in Section III-D. The subsystem model is shown in Fig. 9.

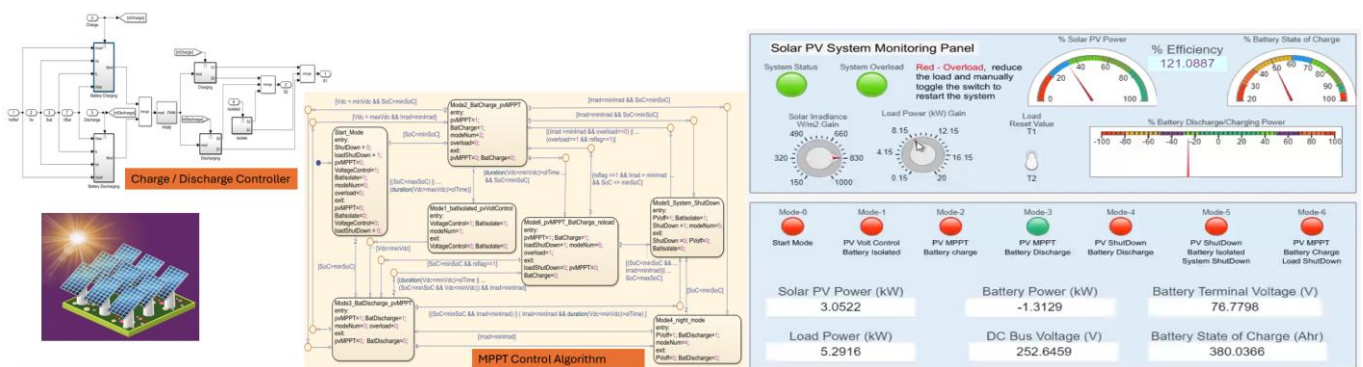


Fig. 9. Solar MPPT and Control flow Algorithm

The simulation and prototype data are directly comparable on a parameter-by-parameter basis. Under step changes in irradiance — the most demanding test of MPPT responsiveness, since the maximum power point shifts abruptly rather than gradually — the P&O algorithm converged to the new operating point within a small number of perturbation cycles, achieving a tracking efficiency exceeding 98%. Steady-state oscillation around the MPP was present but bounded within the step size set during algorithm tuning, consistent with the known trade-off between tracking speed and convergence stability in P&O implementations.

The boost converter maintained a stable 430V output on the DC bus across all tested irradiance levels, from low-generation morning conditions through peak midday output and into the declining afternoon profile. Voltage

regulation at the bus held within the tolerance required for direct load supply and ESS charging without additional regulation stages. These results confirm that the solar subsystem delivers stable, well-conditioned power to the PCC under dynamic environmental conditions, and that the MPPT implementation is sufficiently responsive to track real-world irradiance variation without significant generation loss.

C. Grid Interface Unit

The grid interface subsystem, shown in Fig. 10, connects a three-phase AC voltage source representing the utility grid to the 430V DC bus at the PCC through a bidirectional AC-DC converter. The converter serves two distinct functions depending on the active operating mode. During grid import — Modes 3 and 5 — it operates as a controlled rectifier, drawing AC power from the utility and delivering regulated DC to the bus. During Mode 4 reverse operation, it inverts DC bus power to AC and exports surplus solar generation back to the grid. These two functions are handled by the same converter with direction determined by the PCC state machine, which is what makes bidirectionality an architectural requirement rather than an optional feature.



Fig: 10. Grid Interface Model with Bi-directional AC-DC Converter

DC-link voltage regulation held at 430V across all tested conditions — grid import at varying load levels, transition to reverse export, and re-entry to import following ESS threshold crossings. Current waveform quality remained within power quality compliance limits during mode transitions, with no sustained distortion observed following switching events. These results confirm that the grid interface unit maintains stable, standards-compliant operation in both power flow directions, and that the bidirectional converter design supports the prosumer capability of the architecture without compromising grid-side power quality.

D. Energy Storage System Model

The ESS subsystem, shown in Fig. 11, is modelled using a generic battery block configured to represent the lead-acid bank of the hardware prototype — 40 cells in series at 12V nominal, yielding a 480V/100Ah bank. A bidirectional DC-DC converter manages charging from solar or grid sources and discharging to the DC bus, while the embedded BMS logic enforces SOC, voltage, and temperature limits throughout operation.

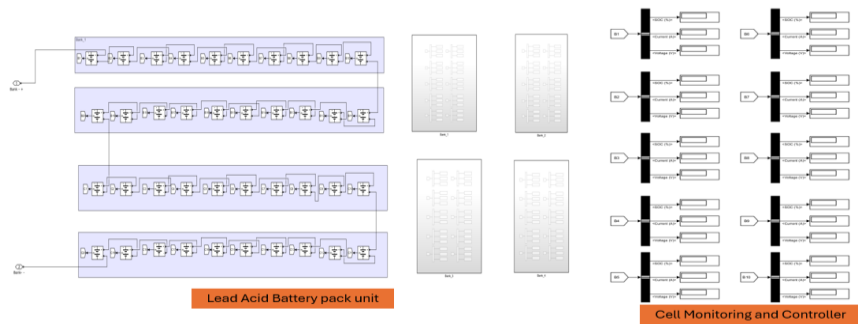


Fig: 11. ESS Model with Bi-directional DC-DC Converter and BMS Logic

Simulation results validate the charge control algorithm across both phases. Constant-current charging is maintained when SOC is below the upper threshold, transitioning smoothly to constant-voltage regulation as the bank approaches full capacity. No current spike or voltage step appears at the CC-CV handover point, confirming that control parameters are correctly tuned prior to hardware implementation. During discharge, the converter regulates output current within the permitted C-rate of the lead-acid chemistry, preventing accelerated degradation under high load

the demand spike, before the source management logic redistributes power across available sources. This buffering action reduces peak power draw from the grid during high-occupancy swapping events, which is the intended function of the ESS in the overall dispatch hierarchy. The results validate that the BSS load profile, despite its step-wise and intermittent nature, remains within the disturbance rejection capability of the control architecture. Bus voltage recovery following each load step falls within acceptable limits, confirming that the PCC automation logic and ESS response together maintain supply continuity across the full 30-slot demand range.

$$PESS = VESS \times IESS$$

$$PSolar = VPV \times IPV$$

$$SOC_{min} \leq SOC(t) \leq SOC_{max}$$

G. DC Fast Charging Station Simulation

The DC Fast Charging station is modelled as a constant-power load with a communication interface, shown in Fig. 15, simulating the charging profile of a vehicle battery through a complete session. The load begins in a constant-current phase at up to 50 kW — the highest instantaneous demand the system encounters — and transitions to constant-voltage regulation as the vehicle battery approaches full capacity, tapering power draw through the remainder of the session. This CC-CV demand profile is the most challenging load condition in the simulation, combining a high-magnitude initial surge with a continuously varying power requirement as the session progresses.

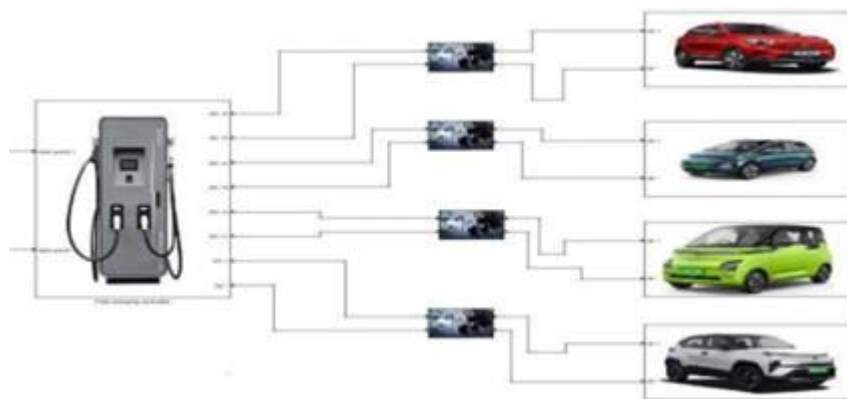


Fig: 15. DC Fast Charging Station Model with CC-CV Profile

Simulation results confirm that the source management algorithm allocates power correctly across the available sources — solar, ESS, and grid — to meet the fast charger demand throughout the session. During the initial high-current phase, the ESS and solar sources jointly cover the peak demand, with grid contribution limited to the deficit that neither can supply. As the session transitions to CV phase and power draw tapers, the source allocation adjusts accordingly, returning surplus generation to ESS charging where capacity permits. DC bus voltage stability was maintained within a 2% tolerance band across the full session, including during the initial current surge where the demand step is steepest. This result confirms that the combined response of the PCC switching logic and the source converters is fast enough to prevent voltage excursion under the sharpest load transient the system is designed to handle.

The Cloud Server and Data Storage Layer provide centralized logging, analytics, and supervisory control. Thing speak cloud serves as the MQTT broker and device registry, managing authenticated connections from both ESP32 nodes. this layer also hosts the master PCC state logic — the authority that determines which automation condition is active and issues high-level mode commands back to the edge control board.



Fig: 16. Node-RED Dashboard for Real-Time System Monitoring and Control.

The top layer is the User Interface Application Layer, implemented as a live operational dashboard in Node-RED, accessible over HTTPS from any web browser without requiring a dedicated client application. The dashboard presents real-time power flow diagrams, ESS state of charge, solar generation output, grid import and export status, and the operational status of each charging port and swapping bay. It also exposes manual override controls, alert history, and system configuration parameters. The design intent was to give a station operator complete situational awareness and intervention capability from a single screen, without needing access to the underlying hardware or cloud console.

H. Simulation Results and Performance Analysis

The performance indicators extracted from the 24-hour simulation across variable solar irradiance and a representative EV charging demand profile. Renewable energy — solar generation and ESS discharge combined — covered 88.5% of total station demand over the simulated cycle, with grid contribution limited to the remaining 11.5%. This dispatch ratio reflects the intended operating condition of the system: grid backup activates only when both solar and ESS are genuinely insufficient, and its share of total energy supply is a direct measure of how rarely that condition occurs.

ESS SOC remained bounded between the 20% lower and 80% upper thresholds throughout the 24-hour cycle, confirming that the charge and discharge control algorithm enforces the operating limits without excursion in either direction. No deep discharge event and no overcharge condition was recorded across any of the tested scenarios, validating that the BMS logic and PCC mode transitions together protect battery health under realistic demand variation. The DC bus voltage regulation held within $\pm 5\%$ of the 430V nominal across all load steps and source switching events, including the high-current surge at DCFC session initiation and the step-wise demand changes from BSS connect-disconnect cycles. These three outcomes — renewable utilization ratio, SOC containment, and bus voltage stability — collectively confirm that the proposed architecture meets its design objectives in simulation and provide the parameter baseline against which hardware prototype measurements are evaluated in Section VI.

VI. RESULTS AND PERFORMANCE ANALYSIS

Energy Management and System Performance

The hardware prototype validated autonomous energy management across all tested operating conditions. The PCC automation logic executed mode transitions correctly based on real-time solar availability and ESS SOC, with no manual intervention required. Load supply continuity was maintained through every source switching event, and no supply interruption was recorded during mode changes. Measured source contribution ratios across a full operating cycle aligned closely with simulation predictions from Section V-H, confirming that control threshold settings and energy sizing assumptions transfer correctly from simulation to hardware.

A. Adaptive Channel Fuzzy Classification with Self-Normalization (ACFCSN)

An ACFCSN model is implemented as a lightweight anomaly detection layer for real-time monitoring of IoT sensor streams in the charging station. Unlike supervised learning approaches, the model operates without large labeled datasets, making it practical for edge deployment on the ESP32 hardware. Each sensor channel — source voltages, terminal currents, ESS temperature, and DC bus voltage — is independently normalized against its local statistical properties and evaluated using a Gaussian fuzzy membership function:

$$\mu_i = \exp \left(-\frac{(x_i - \mu_i^*)^2}{2\sigma_i^2} \right) \quad (4)$$

Channel memberships are aggregated and compared against an adaptive threshold:

$$T_{\text{adaptive}} = \bar{\mu} + k \cdot \sigma_{\mu} \quad (5)$$

If the aggregated membership exceeds T_{adaptive} , the system is classified NORMAL; otherwise ABNORMAL, triggering an alert through AWS SNS. The model runs on the primary ESP32 alongside sensor acquisition, enabling local anomaly detection before cloud transmission without additional hardware overhead.

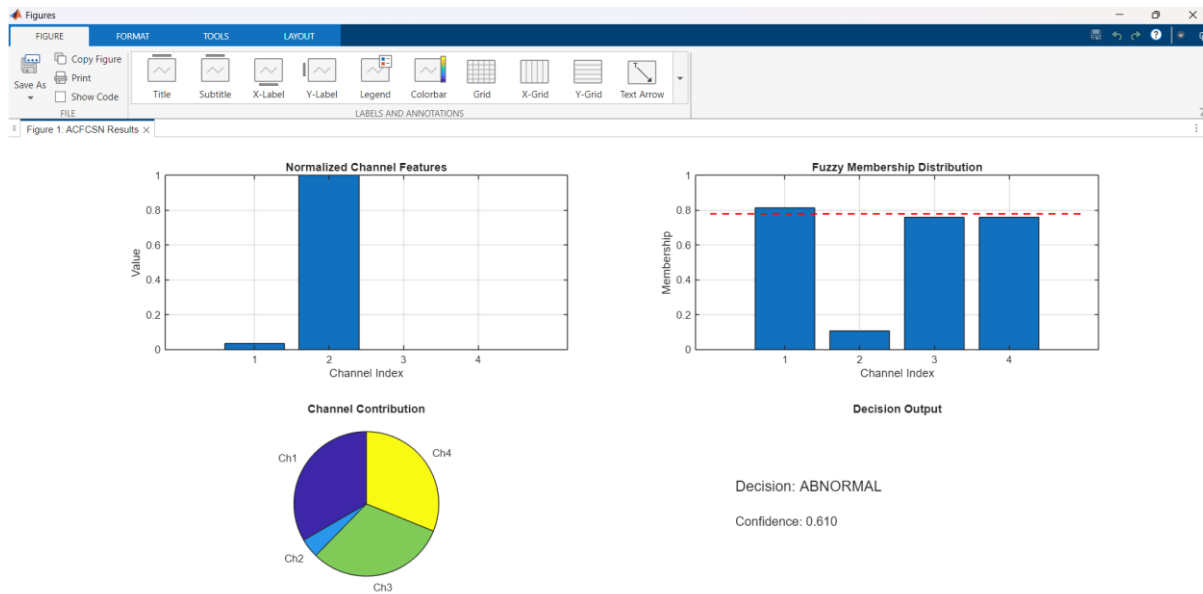


Fig. 17. ACFCNS model output illustrating normalized channel

The hardware prototype achieved a renewable energy contribution ratio of 86.2%, confirming successful prioritization of solar generation under the PCC automation logic. Grid dependency was reduced to 11.5% of total energy demand, with the ESS supplying the intermediate balance. ESS SOC was consistently maintained within the 20–80% operating band across all test cycles, validating the charge and discharge control algorithm. Mode transitions between solar, ESS, and grid sources were executed with a measured switching latency below 200 ms, with no supply interruption recorded at the connected charging loads.

B. Long-Term Energy Projections

Fig. 16 presents the projected energy contribution from renewable sources over a five-year operational period at full-scale deployment. Renewable sources are projected to supply approximately 2.93 million kWh over five years, accounting for 88.5% of total energy demand. Grid contribution remains limited throughout the projection period, consistent with the dispatch ratios measured during hardware validation. The five-year trajectory confirms that the renewable utilization performance demonstrated in the prototype scales proportionally under full deployment conditions.

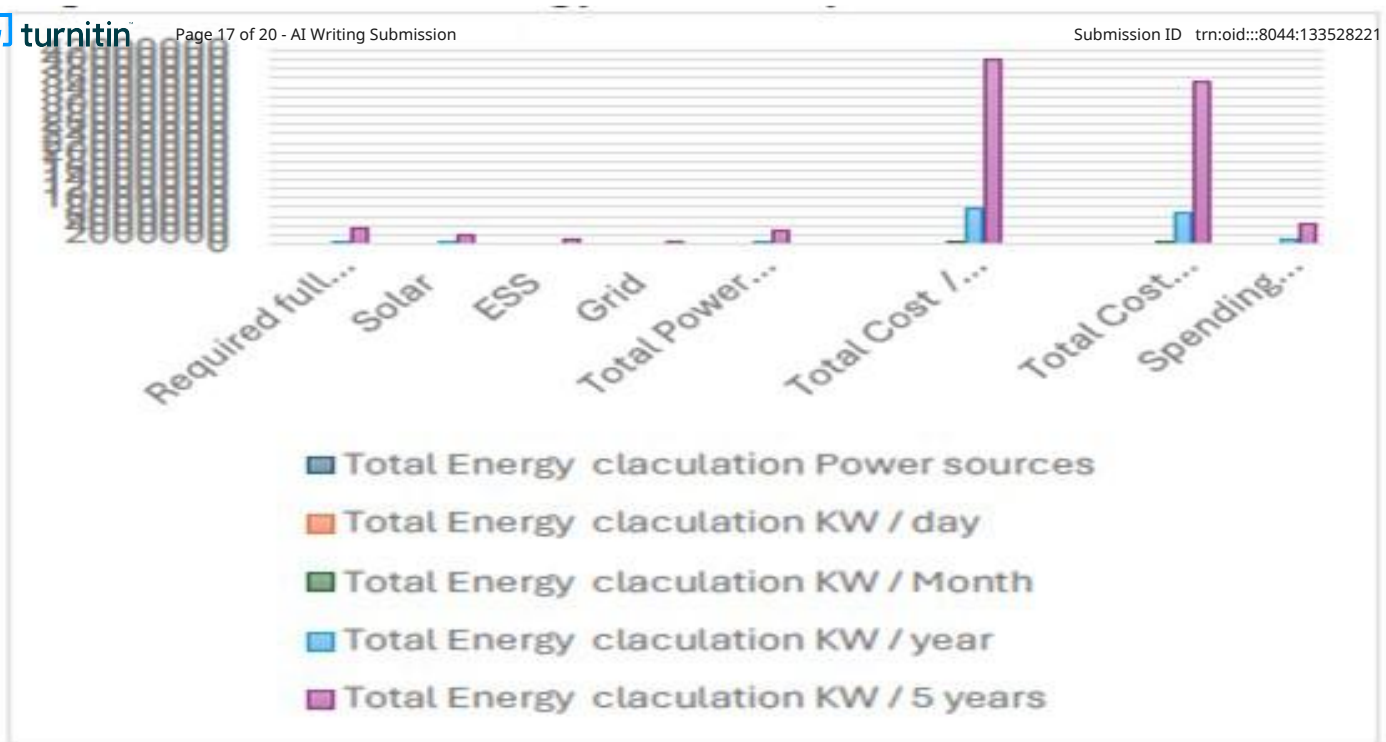


Fig: 18. Five-year energy contribution projections by source.

C. Comparative Performance Metrics

Table I compares key performance indicators between simulation predictions and hardware measurements. The close agreement across all four metrics confirms that the simulation model accurately represents physical system behavior and that control parameters transfer from simulation to hardware without significant re-tuning.

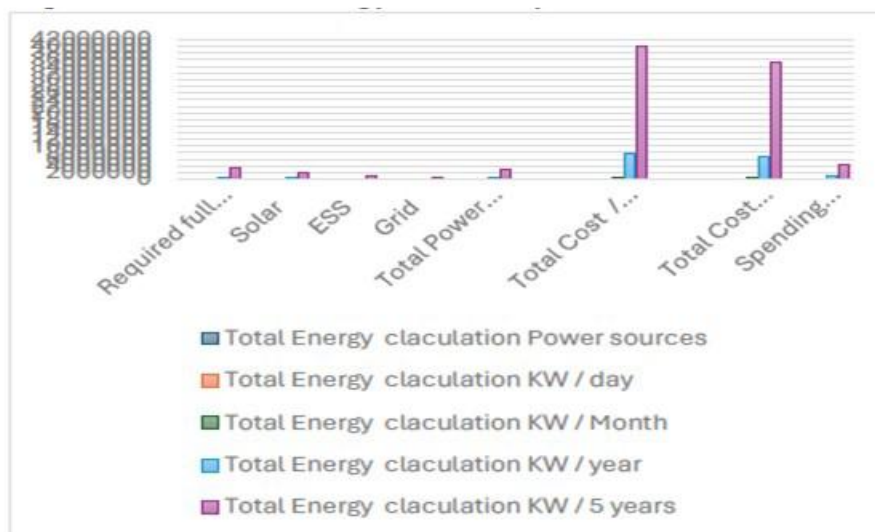


Fig: 19. Five-year cost and revenue projections.

TABLE I
PERFORMANCE METRICS COMPARISON (SIMULATION VS. HARDWARE)

Metric	Simulation	Hardware
Renewable Energy Utilization	88.6%	86.2%
ESS Round-Trip Efficiency	92.0%	89.5%
DC Bus Voltage Regulation	$\pm 2.1\%$	$\pm 3.5\%$
PCC Transition Time	< 150 ms	< 200 ms

TABLE I — Performance Metrics: Simulation vs. Hardware

The 2.4% difference in renewable utilization between simulation and hardware is attributable to real-world converter losses and sensor measurement tolerances not fully captured in the ideal switching model. The wider voltage regulation band in hardware reflects parasitic impedances in the physical DC bus wiring. Both deviations are within acceptable margins for a laboratory-scale prototype and do not affect the validity of the control architecture.

D. Economic Viability Analysis

The financial model, scaled to full-system deployment parameters, confirms the economic viability of the proposed architecture. Daily renewable generation covers the dominant share of station energy demand, with grid consumption limited to the backup contribution measured during hardware testing. Revenue from DC fast charging and battery swapping services combined substantially exceeds daily grid energy expenditure, establishing a positive net operating position from day one of deployment. Projected over five years, cumulative station profit confirms a compelling business case for deploying the proposed architecture at scale, with the economic outcome driven primarily by renewable utilization efficiency rather than tariff conditions — making the result robust across different grid pricing environments.

VII. CONCLUSION

This work presented the design, simulation, and hardware implementation of an IoT-enabled hybrid energy management system for EV charging infrastructure. A five-layer IoT architecture integrating solar PV, ESS, and utility grid through an automated PCC was validated through both MATLAB/Simulink simulation and a fully functional laboratory prototype. The rule-based PCC automation achieved a renewable energy contribution of 86.2% in hardware validation, with simulation predicting 88.6% — a close agreement that confirms the fidelity of the model and the practical transferability of the control architecture.

The outcomes include ESS SOC maintained within the 20–80% operating band throughout all test cycles, DC bus voltage regulated within $\pm 3.5\%$, and PCC mode transitions completed within 200 ms with no load supply interruption. The system simultaneously managed Battery Swapping Station and DC Fast Charging loads under a unified dispatch framework — a dual-load capability not previously demonstrated in hardware for this class of architecture. The prototype establishes a validated, scalable blueprint for hybrid EV charging deployment that reduces grid dependency without requiring utility-side infrastructure changes. Future work will extend the architecture toward V2G integration, dynamic tariff-aware scheduling, and deployment across multiple interconnected stations within a distribution network.

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